

Week 6 - Hash Functions

MAT260: Cryptology

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Introduction:

The assigned computer problems this week are from Chapter 12, Problems 1 and 2.

Make sure to review the example computer problems in "Appendix C.8 Examples for Chapter 12" that work similar problems to those you are assigned.

Make sure to run your code so all relevant computations/results are displayed and then export your work as a PDF file for submission.

Chapter 12 Problems:

Problem 1: In this exercise, I used MATLAB to calculate the probability that at least two people in a room share the same birthday.

For part 1, I computed both the exact probability and an approximation (using formula 8.1) for a group of 30 people,

showing that the chance of a shared birthday is about 70.63%. In part 2, I used the approximation formula in a loop to determine

that at least 58 people are needed for a 99% chance of a birthday match. Finally, for part 3, I applied the pigeonhole principle to

show that with 366 people, a birthday match is guaranteed, since there are only 365 possible birthdays.

```
format long

% Part 1: Probability with 30 people
r = 30;
N = 365;
prob_no_match = prod(1 - (0:r-1)/N);
prob_match = 1 - prob_no_match;
fprintf('Probability at least two of 30 people share a birthday: %.6f\n',
prob_match);
```

Probability at least two of 30 people share a birthday: 0.706316

```
% Approximation using Formula (8.1)
approx_match = 1 - exp(-r^2 / (2 * N));
fprintf('Approximate probability (Formula 8.1): %.6f\n', approx_match);
```

Approximate probability (Formula 8.1): 0.708547

```
% Part 2: Minimum r for 99% probability
target = 0.99;
r = 1;
while 1 - exp(-r^2 / (2 * N)) < target
    r = r + 1;
end
fprintf('Minimum number of people for 99% match probability: %d\n', r);
```

Minimum number of people for 99% match probability: 58

```
% Part 3: 100% probability
fprintf('Number of people for 100% match (pigeonhole principle): 366\n');
```

Number of people for 100% match (pigeonhole principle): 366

Problem 2:

To solve this problem, I treated it as a variation of the birthday problem, where the total number of possible outcomes is 10,000 (since a 4-digit SSN ranges from 0000 to 9999).

Using MATLAB, I calculated the probability that at least two students in a class of 200 share the same 4 digit SSN by first computing the probability that all students have unique numbers,

and then subtracting that from 1. I also used the exponential approximation formula from the textbook to verify the result. Both methods confirmed that there's a high probability

of a collision demonstrating why using only the last four digits of SSNs is not secure for student identification.

```
format long
N = 10000; % 4-digit SSNs
r = 200;
prob_no_match = prod(1 - (0:r-1)/N);
prob_match = 1 - prob_no_match;
fprintf('Probability at least two students share 4-digit SSN: %.6f\n', prob_match);
```

Probability at least two students share 4-digit SSN: 0.865120

